PROJECT SYNOPSIS

Interpretable ECG-based Apnea Detection: HRV, EDR, ECG Morphology and Signal-Quality Analysis with QRS-Error, Window-Selection and Patient-Independent Evaluation

Course	**Digital Signal Processing (ECN15101)**
Project Area	**Biomedical / Wearable Signal Processing**

Project Members

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1. Project Overview

Sleep apnea is characterized by repeated episodes of reduced or temporarily absent breathing during sleep. Although respiration is traditionally monitored directly, the electrocardiogram (ECG) also contains indirect information related to respiratory activity and autonomic nervous-system changes. These effects can appear through beat-to-beat heart-rate variation, RR-interval dynamics and changes in ECG morphology. This project, undertaken as part of the Digital Signal Processing course (ECN15101), aims to investigate how digital signal-processing techniques can extract these patterns from ECG recordings and use them to identify segments associated with apnea-like events.

2. Primary Objective

To design and evaluate a DSP-based pipeline that processes overnight ECG recordings, detects cardiac beats, constructs RR-interval and heart-rate-variability (HRV) representations, extracts ECG-derived respiratory information, and classifies short time windows as normal or apnea-like using the available annotations as ground truth.

3. Dataset and Data Source

Dataset: PhysioNet Apnea-ECG Database, Version 1.0.0.

Official source: https://physionet.org/content/apnea-ecg/1.0.0/

- The database contains 70 overnight ECG records, divided into 35 learning records and 35 test records.
- Recordings range from slightly less than 7 hours to nearly 10 hours, making the dataset suitable for studying long-duration sleep-related signal patterns.
- The ECG is digitized at 100 samples/second using 16-bit samples.
- The learning records contain apnea annotations at one-minute intervals indicating the presence or absence of apnea during each minute.
- The records also provide machine-generated QRS annotations; these will be treated cautiously because the PhysioNet documentation notes that the supplied QRS annotations are unaudited.
- A subset of eight recordings additionally contains respiratory-effort, airflow and SpO_2 signals. These signals can be used for validation or comparison of ECG-derived respiratory information where appropriate.

Dataset reference: Penzel, T., Moody, G. B., Mark, R. G., Goldberger, A. L., & Peter, J. H. "The Apnea-ECG Database," Computers in Cardiology, 2000.

4. Proposed DSP Methodology / Procedure of Action

Dataset acquisition and record selection: Download the Apnea-ECG Database from PhysioNet and select representative learning-set records containing ECG signals and minute-level apnea annotations. The learning set will primarily be used for development, feature analysis and model/threshold selection.

Signal loading and initial inspection: Read the ECG signal and corresponding header/annotation information. Plot representative portions of raw overnight ECG recordings to understand amplitude range, baseline drift, noise and beat morphology.

Pre-processing and digital filtering: Remove baseline wander using an appropriate high-pass or band-pass filtering stage and attenuate high-frequency noise using a suitable low-pass stage. If power-line interference is observed, apply a narrow notch filter around the relevant mains frequency. Filter parameters will be selected so that the QRS morphology is preserved.

R-peak / QRS detection: Detect R-peaks using a suitable QRS detection algorithm. The detected beat locations will be checked visually against the ECG, and obvious false or missed detections will be handled during preprocessing/quality control.

RR-interval time series construction: Compute successive RR intervals from the detected R-peak positions. Convert the beat-to-beat sequence into a time series suitable for HRV and frequency-domain analysis, including appropriate handling of ectopic or abnormal intervals where necessary.

Heart-rate variability analysis: Calculate time-domain HRV features such as mean RR, standard deviation of RR intervals and related variability measures. Investigate how these features differ between normal and apnea-associated windows.

Frequency-domain analysis: Transform the processed RR/HRV series into the frequency domain using FFT and/or PSD estimation. Analyze the distribution of spectral energy and identify frequency-domain characteristics that may distinguish normal and apnea-like periods.

ECG-derived respiratory signal extraction: Investigate respiration-related modulation embedded in the ECG. Candidate approaches include respiratory information derived from beat-to-beat interval variation and/or ECG amplitude or morphology variation. The extracted respiratory signal will be compared with the available respiration channels for the subset of records where those channels exist.

Windowing and feature extraction: Divide the overnight recordings into fixed-length windows, with the window size selected based on the temporal scale of apnea events. For every window, calculate a feature vector combining RR/HRV, spectral and respiratory-related features.

Normal vs. apnea-like classification: Map each feature window to the corresponding minute-level apnea annotation. Start with interpretable threshold-based rules and then evaluate a simple machine-learning classifier such as logistic regression, decision tree or random forest if justified by the results.

Performance evaluation: Evaluate the system using suitable metrics such as accuracy, precision, recall/sensitivity, specificity, F1-score and confusion matrix. Results will be interpreted at the window/minute level, and where possible the effect of different window sizes and feature groups will be compared.

Visualization and final analysis: Present the complete processing chain through plots of raw and filtered ECG, detected R-peaks, RR tachogram, HRV spectrum/PSD, ECG-derived respiration and predicted versus annotated apnea periods. Discuss which DSP stages and features contribute most to the final detection performance.

5. Proposed Processing Pipeline

Raw overnight ECG
→ Noise / baseline-wander removal
→ Optional power-line notch filtering
→ QRS / R-peak detection
→ RR-interval time series
→ HRV time-domain + frequency-domain analysis
→ ECG-derived respiratory information
→ Window-level feature extraction
→ Normal / apnea-like classification
→ Performance evaluation against annotations

6. Expected Outputs

- Cleaned and filtered ECG waveforms with representative before/after comparisons.
- Accurate R-peak locations and RR-interval time series.
- RR tachograms and heart-rate / HRV measures for normal and apnea-associated windows.
- FFT/PSD-based frequency-domain characterization of HRV.
- ECG-derived respiratory signal and comparison with directly recorded respiration where available.
- Window-level feature dataset linking DSP-derived features to apnea annotations.
- Normal versus apnea-like classification results with quantitative performance metrics.
- Plots and visualizations showing agreement/disagreement between detected events and reference annotations.

7. Scope and Research Boundaries

- The project is a Digital Signal Processing research/academic prototype and is not intended to function as a clinical diagnostic system.
- The primary focus is on signal acquisition from the dataset, digital filtering, feature extraction, signal characterization and event classification.
- The project will emphasize interpretable DSP concepts—filtering, QRS detection, RR analysis, HRV, FFT/PSD and ECG-derived respiration—rather than relying exclusively on a black-box deep-learning model.
- Reference annotations from the dataset will be used for evaluation, while the supplied QRS annotations may be cross-checked because they are explicitly described by PhysioNet as unaudited.

8. Possible Extensions

- Compare multiple ECG-derived respiration techniques and quantify their agreement with the available respiration channels.
- Study the effect of different analysis-window lengths and overlap ratios.
- Compare time-domain, frequency-domain and combined feature sets.
- Investigate alternative QRS/R-peak detection algorithms and their effect on downstream HRV features.
- Compare threshold-based detection with lightweight machine-learning approaches.
- Evaluate subject-independent performance by keeping complete subjects/records separate between development and final testing.
- Explore real-time or wearable-device implementation as a future extension.

9. Deliverables

- Working ECG preprocessing and filtering module.
- R-peak detection and RR-interval generation module.
- HRV time-domain and frequency-domain analysis module.
- ECG-derived respiration extraction module.
- Feature extraction and apnea-like event classification module.
- Final plots, quantitative evaluation and comparison of methods.
- Project report documenting DSP theory, implementation, observations and conclusions.

10. Recent Related Research and References

Recent studies from 2024–2025 directly overlap with important parts of the proposed methodology. They provide the research basis for the project while also showing how our approach differs from more advanced AI-based methods.

10.1 Parbat & Chakraborty (2024) — "Multiscale entropy analysis of single lead ECG and ECG derived respiration for AI based prediction of sleep apnea events." Biomedical Signal Processing and Control, 87 (2024), 105444. DOI: 10.1016/j.bspc.2023.105444.

Similarity: Uses single-lead ECG, heart-rate information and ECG-derived respiration (EDR) to obtain features for sleep-apnea event prediction.

Difference: Its main approach is multiscale entropy followed by AI models. Our project emphasizes conventional DSP—filtering, QRS/R-peak detection, RR/HRV, FFT/PSD and EDR—followed by interpretable thresholding or lightweight ML.

10.2 Tyagi & Agrawal (2024) — "Automatic detection of sleep apnea from a single-lead ECG signal based on spiking neural network model." Computers in Biology and Medicine, 179 (2024), 108877. DOI: 10.1016/j.compbiomed.2024.108877.

Similarity: Uses the same PhysioNet Apnea-ECG database, extracts HRV and EDR from 1-minute ECG segments, and performs apnea-event classification.

Difference: It uses a spiking neural network with temporal encoding. Our project keeps the emphasis on the DSP processing chain and starts with interpretable features and simple classifiers such as logistic regression, decision tree or random forest.

10.3 Yamane, Fujii & Morita (2025) — "Clinical-level screening of sleep apnea syndrome with single-lead ECG alone is achievable using machine learning with appropriate time windows." Sleep and Breathing, 29 (2025), Article 156. DOI: 10.1007/s11325-025-03316-0.

Similarity: Uses the same 70-record PhysioNet Apnea-ECG database, single-lead ECG, R-peak/RR information, frequency-domain features using Welch PSD with FFT, and ML classification.

Difference: Its central contribution is optimizing past/future ECG information around a target 1-minute apnea epoch and comparing several ML classifiers. Our project focuses more strongly on the underlying DSP stages and the contribution of HRV and EDR; window-size comparison can be an extension.

10.4 Comparison with the Proposed Project

Reference	Main similarity	Main difference	What we take from it
Parbat & Chakraborty (2024)	ECG + HR + EDR for apnea prediction	Multiscale entropy + AI	Supports EDR and feature-based ECG apnea analysis.
Tyagi & Agrawal (2024)	Same dataset; HRV + EDR; 1-min segments	Spiking neural network	Strongest reference for the overall signal-to-classification pipeline.
Yamane et al. (2025)	Same dataset; RR/R-peak features; FFT/Welch PSD; ML	Past/future window optimization	Supports spectral features and motivates window-size experiments.
Our project	Filtering + R-peaks + RR/HRV + FFT/PSD + EDR + classification	DSP-first, interpretable, lightweight	Evaluate how conventional DSP features distinguish apnea-like windows.

10.5 Bibliographic References

1. Parbat, D., & Chakraborty, M. (2024). Multiscale entropy analysis of single lead ECG and ECG derived respiration for AI based prediction of sleep apnea events. Biomedical Signal Processing and Control, 87, 105444. DOI
2. Tyagi, P. K., & Agrawal, D. (2024). Automatic detection of sleep apnea from a single-lead ECG signal based on spiking neural network model. Computers in Biology and Medicine, 179, 108877. DOI
3. Yamane, T., Fujii, M., & Morita, M. (2025). Clinical-level screening of sleep apnea syndrome with single-lead ECG alone is achievable using machine learning with appropriate time windows. Sleep and Breathing, 29, 156. DOI